Diesel Generator Set

mtu 12V4000 DS2250

380V – 11 kV/50 Hz/standby power/NEA (ORDE) + Tier 2 optimized 12V4000G94F/water charge air cooling

Optional equipment and finishing shown. Standard may vary.

Product highlights

Benefits
- Low fuel consumption
- Optimized system integration ability
- High reliability
- High availability of power
- Long maintenance intervals

Support
- Global product support offered

Standards
- Engine-generator set is designed and manufactured in facilities certified to standards ISO 2008:9001 and ISO 2004:14001
- Generator set complies to ISO 8528
- Generator meets NEMA MG1, BS 5000, ISO, DIN EN and IEC standards
- NFPA 110

Power rating
- System ratings: 2220 kVA - 2300 kVA
- Accepts rated load in one step per NFPA 110*
- Generator set complies to G3 according to ISO 8528-5
- Generator set exceeds load steps according to ISO 8528-5*

Performance assurance certification (PAC)
- Engine-generator set tested to ISO 8528-5 for transient response
- 85% load factor
- Verified product design, quality and performance integrity
- All engine systems are prototype and factory tested

Complete range of accessories available
- Control panel
- Power panel
- Circuit breaker/power distribution
- Fuel system
- Fuel connections with shut-off valve mounted to base frame
- Starting/charging system
- Exhaust system
- Mechanical and electrical driven radiators
- Medium and oversized voltage alternators

Emissions
- Tier 2 optimized engine
- NEA (ORDE) optimized engine

Certifications
- CE certification option
- Unit certificate acc. to VDE-AR-N 4110

* Changes to the standard parameter sets (alternator-regulator and genset-controller) are necessary

A Rolls-Royce solution

Application data [1]

Engine

Manufacturer	*mtu*
Model	12V4000G94F
Type	4-cycle
Arrangement	12V
Displacement: l	57.2
Bore: mm	170
Stroke: mm	210
Compression ratio	16.4
Rated speed: rpm	1500
Engine governor	ADEC (ECU 9)
Max power: kWm	1930
Air cleaner	dry

Fuel system

Maximum fuel lift: m	5
Total fuel flow: l/min	27

Fuel consumption [2]

	l/hr	g/kwh
At 100% of power rating:	463	199
At 75% of power rating:	360	206
At 50% of power rating:	249	214

Liquid capacity (lubrication)

Total oil system capacity: l	260
Engine jacket water capacity: l	160
Intercooler coolant capacity: l	40

Combustion air requirements

Combustion air volume: m^3/s	2.4
Max. air intake restriction: mbar	50

Cooling/radiator system

Coolant flow rate (HT circuit): m^3/hr	55
Coolant flow rate (LT circuit): m^3/hr	30
Heat rejection to coolant: kW	790
Heat radiated to charge air cooling: kW	480
Heat radiated to ambient: kW	75
Fan power for electr. radiator (40°C): kW	55

Exhaust system

Exhaust gas temp. (after engine): °C	460
Exhaust gas temp., max (after engine): °C	550
Exhaust gas temp. (before turbocharger): °C	700
Exhaust gas volume: m^3/s	6.2
Maximum allowable back pressure: mbar	50

Standard and optional features

System ratings (kW/kVA)

Generator model	Voltage	NEA (ORDE) + Tier 2 optimized					
		without radiator			with mechanical radiator		
		kWel	kVA*	AMPS	kWel	kVA*	AMPS
Leroy Somer LSA52.3 S7 (Low voltage Leroy Somer standard)	380 V	1840	2300	3494	1784	2230	3388
	400 V	1840	2300	3320	1784	2230	3219
	415 V	1840	2300	3200	1784	2230	3102
Leroy Somer LSA52.3 L12 (Low voltage Leroy Somer oversized)	380 V	1840	2300	3494	1784	2230	3388
	400 V	1840	2300	3320	1784	2230	3219
	415 V	1840	2300	3200	1784	2230	3102
Leroy Somer LSA53.2 XL9 (Medium volt. Leroy Somer)	11 kV	1840	2300	121	1792	2240	118
Marathon 744RSL7092 (Low voltage Marathon)	380 V	1824	2280	3464	1776	2220	3373
	400 V	1824	2280	3291	1776	2220	3204
	415 V	1808	2260	3434	1776	2220	3088
Marathon 1020FDH7097 (Medium volt. Marathon)	11 kV	1824	2280	120	1776	2220	117

* cos phi = 0.8

1 All data refers only to the engine and is based on ISO standard conditions (25°C and 100m above sea level).

2 Values referenced are in accordance with ISO 3046-1. Conversion calculated with fuel density of 0.83 g/ml. All fuel consumption values refer to rated engine power.